

INFUSION F and O Screener Dashboard

Detailed Project Summary - v4.1.0 heatmap-long-screener

Prepared for options-first Indian F and O stock screening, Upstox-based live data, visual conviction building, manual risk control, and future paper-tested automation readiness.

Current version	v4.1.0-heatmap-long-screener
Capital profile	Rs.50,000 manual morning capital input
Daily loss guard	Rs.2,500 maximum daily loss, equal to 5 percent of capital
Trading mode	Stock option buying focus: CE/PE, monthly stock options preferred
Execution policy	No live auto-orders in Phase 1; paper-first until backtested and proven
Broker/data dependency	Upstox-first migration; Kite removed from active dependency path

Key Objective

The dashboard is being shaped into a practical F and O stock options screener. The purpose is not to show decorative scores, but to teach and convince the trader whether a stock is actually moving, where it becomes positive or negative, what the option bias should be, and whether the risk-to-reward is acceptable before trade execution.

Executive Summary

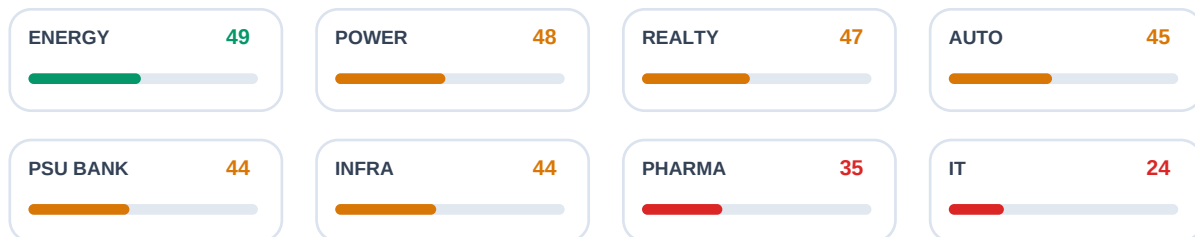
- Upstox is now the primary broker/data dependency.
- The UI direction is light mode, long-form, and screener-first rather than a cramped dark terminal.
- Strength and Conviction are first-class sortable scanner columns.
- Market Pulse now includes Sector Heatmap, Stock Heatmap, Top Gainers, and Top Losers tabs.
- Risk defaults match the approved workflow: Rs.50,000 capital, Rs.2,500 daily loss guard, Rs.500 normal risk, and Rs.750 high-conviction risk.
- Live auto-order execution remains disabled until backtesting and paper trading prove the engine.

1. Current Dashboard Layout

The v4.1 UI is intentionally allowed to scroll vertically. This avoids the previous problem where every panel was forced into one cramped viewport.

Top-to-bottom layout

- Header: Infusion branding, market session, regime, live status, clock, and health dots.
- Market tape: NIFTY, BANKNIFTY, selected symbol, market breadth, leader and laggard sectors.
- Quick controls: Live Data, Scanner, Signals, Watchlist, Alerts, Telegram, and breakout-height control.
- Risk console: Manual capital, daily max loss, risk per trade percentage, normal and high-conviction risk amounts.
- Market Pulse: tabbed sector and stock heatmap section.
- Main screener: scanner table and selected-symbol conviction panel.
- Trade workbench: breakout watch, triggers, news, alert log, and diagnostics.
- Chart popup: selected stock chart appears separately instead of permanently consuming table space.



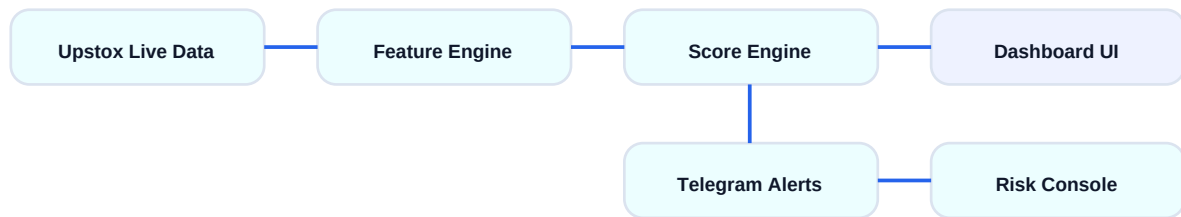
Illustrative v4.1 heatmap style: sector rank, strength score, color intensity, and click-to-filter behavior.

Design direction

- Keep light mode as the default visual language.
- Use green for buy/CE strength, yellow or orange for wait/hold, and red for sell/PE pressure or rejection.
- Keep table columns wide enough for Strength, Conviction, trigger prices, and sector names.
- Use tabs where information is important but not always needed at the same time.

2. Data Flow and Architecture

The dashboard separates broker data ingestion, feature computation, scoring, display, alerts, and risk controls.



Data ingestion	Upstox live feed populates Redis tick and feature state. Kite is no longer expected to be available.
Feature state	Feature hashes include LTP, VWAP, previous close, day high/low, EMA 5/9/20/50, ATR, RSI, MACD, Bollinger compression, relative volume, spread, candle and squeeze labels.
API layer	Dashboard reads endpoints such as /api/ticks, /api/sectors, /api/risk/settings, /api/news/market, /api/options/summary, and alert/analytics endpoints.
Dashboard UI	JavaScript modules render scanner, market pulse, selected-stock insight, option cockpit, trigger panel, news panel, alert log, chart popup, and footer health.
Telegram alerts	Alerts should increasingly use the upgraded signal contract: entry, SL, targets, reason, rejection, and option contract.
AI advisory	ChatGPT API can explain scanner evidence but must not create unsupported signals or invent facts.

Important principle

The dashboard should combine price action, trend alignment, volume expansion, option-chain readiness, sector strength, anti-chase filters, and risk control into a single explainable workflow.

3. Scanner Logic and Visible Columns

The scanner is the main working surface. It should show where a stock is strong, where it is weak, what the option bias is, and whether the setup is tradeable or only watchable.

Symbol / Sector	Stock name and sector grouping for quick filtering.
LTP / Prev close +/-	Live traded price, previous close, point difference, and percentage difference.
RVol	Relative volume expansion versus baseline. Strong movement without volume is treated cautiously.
Trend	BUY, SELL, or HOLD based on VWAP, EMA, MACD, ATR trail, RSI, and price direction evidence.
MTF	Approximate 1M, 5M, 15M, 1H, 4H, 1D dots. Current Phase 1 is proxy-based; true historical MTF is recommended next.
Strength	Setup strength score. This is now wide and sortable.
Conviction	Option-readiness/conviction score. This is now wide and sortable.
Bias	BUY CE, BUY PE, HOLD, or AVOID.
Bull above / Bear below	Spot-price levels where the stock becomes more positive or negative.
Entry / SL / Targets	Phase 1 live proxy trade map. Must be visually confirmed before trade.
Status / Alignment	Armed, watch-only, wait-retest, no-trade, CE aligned, PE aligned, or mixed.

Sorting recommendation

- Default sort should remain Conviction descending during active scanning.
- Use Strength descending when searching for setup formation before signal readiness.
- Use Chg% ascending/descending to detect CE momentum or PE pressure.
- Use RVol descending to identify fresh activity expansion.
- Use Status or Bias filters to separate CE-ready, PE-ready, watch, and avoid groups.

4. Conviction, Strength, and Anti-Chase System

A high score is not enough by itself; the dashboard must show what supports the idea and what is still lacking.

Current evidence stack

- VWAP state: above, below, or not ready.
- EMA state: bullish stack, bearish stack, or mixed.
- MACD state: bullish, bearish, or flat momentum.
- ATR trail: directional confirmation and stop-reference behavior.
- RSI: healthy bullish zone, bearish support zone, overbought, oversold, or neutral.
- Relative volume: whether volume expansion confirms movement.
- Bollinger compression and narrow-range labels: pre-breakout pressure.
- Sector strength: whether stock direction is supported by sector context.
- Spread and liquidity: whether option trading is practical.
- Candle/squeeze patterns: bullish/bearish or compression evidence.

Anti-chase rejection examples

- VWAP stretch too large in ATR terms.
- Signal candle already too large relative to ATR.
- RSI too overheated for fresh CE entry.
- RSI too oversold for fresh PE entry.
- Weak volume relative to move.
- Sector is weak against the intended direction.
- Spread or liquidity is not option-friendly.

Interpretation rule

A trade should be considered only when the dashboard can explain both sides: why the setup has strength and what is still lacking.

5. Options-first Workflow

The user primarily trades stock option buying. Therefore the dashboard should judge the underlying stock first, then use option-chain data to validate whether the CE/PE contract is tradeable.

CE candidate	Requires bullish underlying trend, price reclaim/hold above VWAP or trigger level, healthy RSI, volume expansion, sector support, and acceptable spread.
PE candidate	Requires bearish underlying trend, breakdown/rejection below key level, RSI supporting downside, volume expansion, weak sector/stock pressure, and acceptable spread.
Monthly stock options	Preferred by the user; expiry selection should later consider days to expiry, liquidity, bid-ask spread, IV, and strike distance.
Option chain score	Should combine underlying trend, chain readiness, OI/IV behavior, bid-ask spread, strike liquidity, and premium behavior.
No auto-orders	The dashboard may generate ideas, alerts, and trade maps, but live order placement remains disabled until proven.

Recommended option panel upgrades

- Add OI Spurt and OI Change columns when Upstox option-chain data is stable.
- Show recommended strike, premium, lot affordability, max quantity by risk, IV warning, and spread warning.
- Add expiry suitability: monthly/weekly, days to expiry, theta-risk note, and liquidity rating.
- Separate CE watch, PE watch, CE confirmed, PE confirmed, and rejected setup lists.

6. Risk Management Design

Risk control is a core module. The current settings match the approved manual workflow.

Capital	Rs.50,000 manually updated every morning.
Maximum daily loss	Rs.2,500, equal to 5 percent of capital.
Normal risk per trade	1 percent of capital, currently Rs.500.
High-conviction risk	1.5 percent of capital, currently Rs.750. This should require cleaner alignment and no chase warnings.
Loss day trade cap	2 trades after negative/loss behavior is preferred. Strong discipline avoids revenge trades.
Execution mode	Paper-first; live auto-orders disabled.

Risk recommendation

Although the maximum daily loss is 5 percent, ordinary per-trade risk should remain near 1 percent. High-conviction risk can be used only when trend, sector, volume, option-chain, and anti-chase filters are clean.

7. Alerts, News, and External Context

Alerts and news should support the trader without replacing deterministic scanner logic.

Recommended Telegram alert format

- Symbol, sector, CE/PE bias, timeframe alignment, and score.
- Entry, stop loss, target 1, target 2, and risk-reward.
- Suggested option contract, strike, expiry, premium, spread, IV, OI status, and liquidity rating.
- Why trade: top 3 supporting reasons.
- Why wait/reject: top anti-chase or risk reasons.
- Trade status: WATCH, ARMED, RETEST WATCH, CONFIRMED, NO TRADE, or RISK LOCK.

News source	GDELT first; Google News RSS fallback when GDELT times out or rate-limits.
Use case	Context only: earnings, management news, sector news, regulatory actions, global events, and sudden sentiment shifts.
Limitation	Free feeds are not low-latency newswires. They can be delayed, noisy, duplicated, or rate-limited.
AI advisory	ChatGPT can summarize fetched news and scanner facts only. It must not invent current news or prices.

External inspiration

TradingPulse features page was reviewed for broad UI/workflow inspiration such as sector heatmap, top gainers/losers, OI spurt, scanner conditions, and RR/position sizing: <https://tradingpulse.in/features.html>

8. Known Limitations and Next Build Suggestions

The next phases should prove the engine and make the UI easier to act on.

- True historical MTF is not complete. Current MTF dots are proxy-based from available intraday feature state. Next phase should compute true 1M, 5M, 15M, 1H, 4H, and 1D candles from Upstox historical/live data.
- Option-chain scoring needs deeper live chain ingestion: OI change, IV, bid-ask spread, volume, strike distance, lot affordability, and expiry suitability.
- Backtesting is mandatory before trusting the score engine. Signal counts, suppressed reasons, win rate, target hit, stop hit, and average R:R must be measured.
- The dashboard should separate Watch, Armed, Confirmed, Rejected, and Risk Lock states more visibly.
- Chart popup can be improved into a smoother slide-over with TradingView button, internal chart, and selected trade map.
- Telegram alerts need a trader-friendly format with entry, SL, targets, option contract, and reasons.
- AI should remain advisory-only until deterministic logic is proven.

Suggested development phases

Phase 2	True historical MTF engine from Upstox candles; align 1M/5M/15M/1H/4H/1D with clear CE/PE state.
Phase 3	Option-chain engine: OI, IV, spread, strike selection, expiry suitability, premium confirmation.
Phase 4	Backtesting and paper-trade journal: prove win rate, R:R, drawdown, false positives, and best market regime.
Phase 5	Telegram and dashboard execution assistant: detailed alerts, trade checklist, risk-based quantity, no live orders yet.
Phase 6	Conditional semi-auto execution only after enough proof and manual approval controls.

Bottom line

Infusion should become a conviction-building screener, not a blind signal machine. The best version will show exactly why a stock is moving, where the trade becomes valid, why it may fail, which option contract is practical, and how much risk is allowed before action.